

TrafficVision AI

Smart Traffic Prediction & Congestion Management System



Documentation of Milestone 2

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MILESTONE 2 OBJECTIVE



- **Train traffic prediction models.**
- **Implement congestion forecasting workflows.**
- **Generate traffic prediction reports.**
- **Integrate maps and traffic APIs.**
- **Build alternate route recommendation workflows.**
- **Develop travel time estimation features**



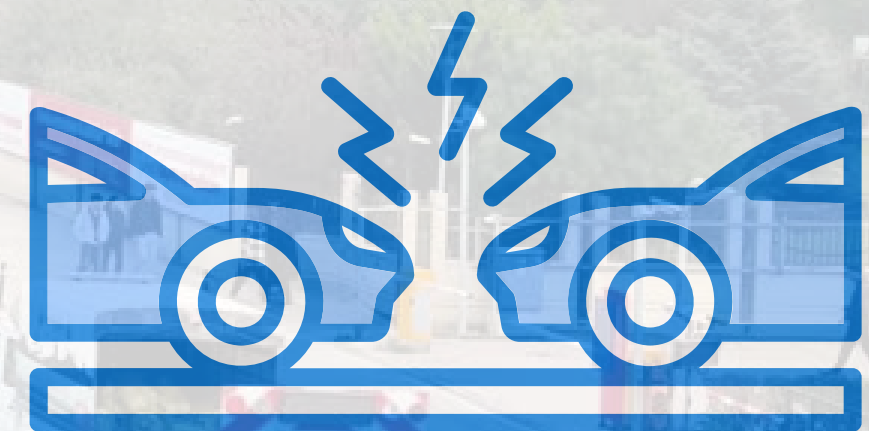
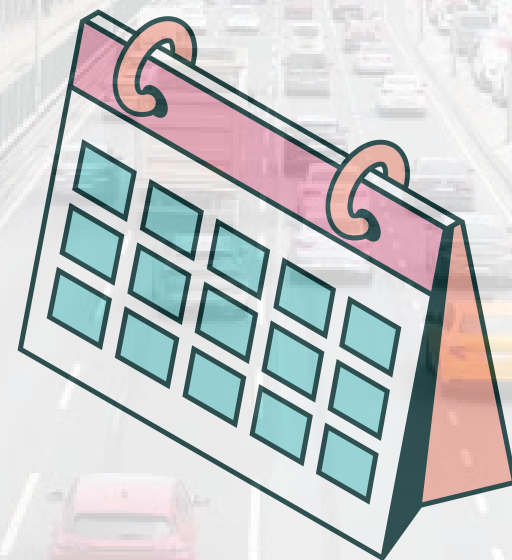
FEATURE SELECTION

Feature selection is the process of selecting the relevant features from the traffic dataset that are useful for predicting congestion.

Selected features in our project:

- Traffic Volume
- Average Speed
- Hour
- Day of Week
- City Zone
- Road Type
- Weather
- Accident Information

Target variable: congestion_level

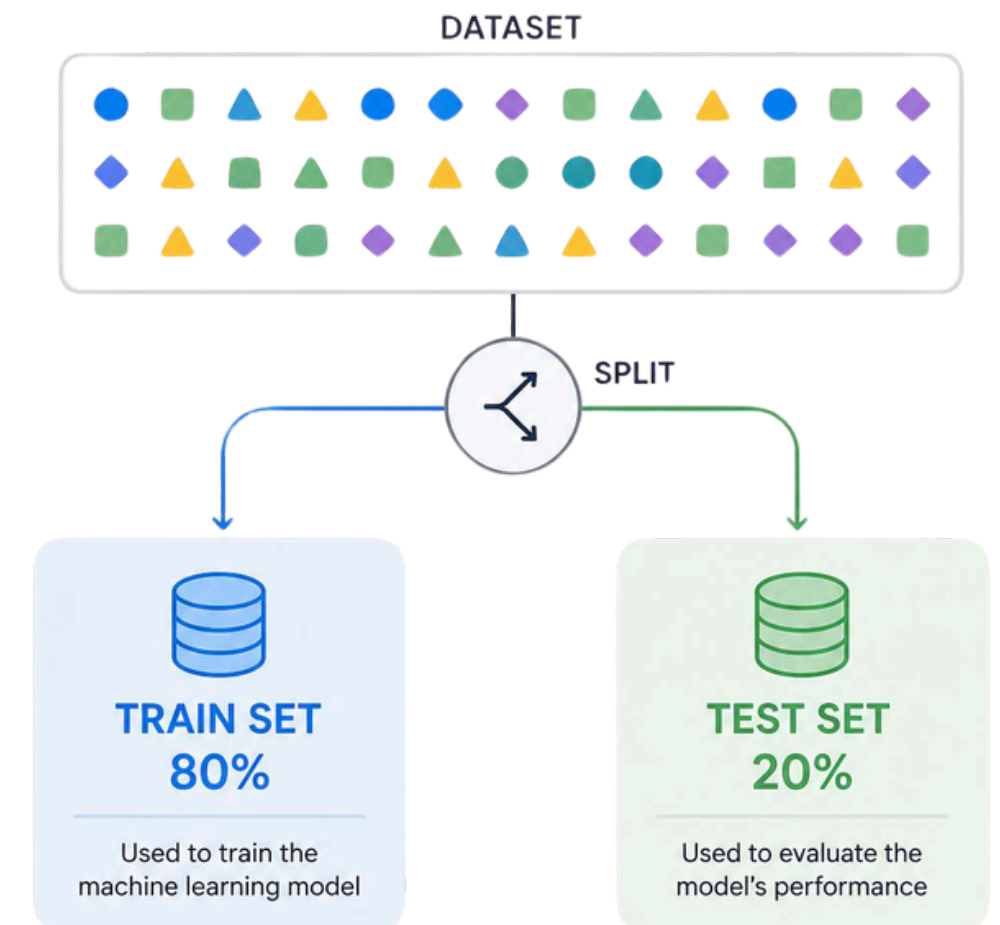


TRAIN-TEST SPLITTING

After selecting the features, the dataset is divided into training data and testing data.

- 80% → Training data
- 20% → Testing data

Training data is used to make the model learn patterns, while testing data is used to check how well the model performs on unseen data.



MODEL TRAINING

- For traffic congestion prediction, we use a Random Forest Classifier.
- The model is trained using the training dataset.
- Random Forest uses multiple decision trees to learn patterns from the given traffic features and make congestion predictions.
- Our model uses 200 estimators.

ALTERNATE ROUTE RECOMMENDATION

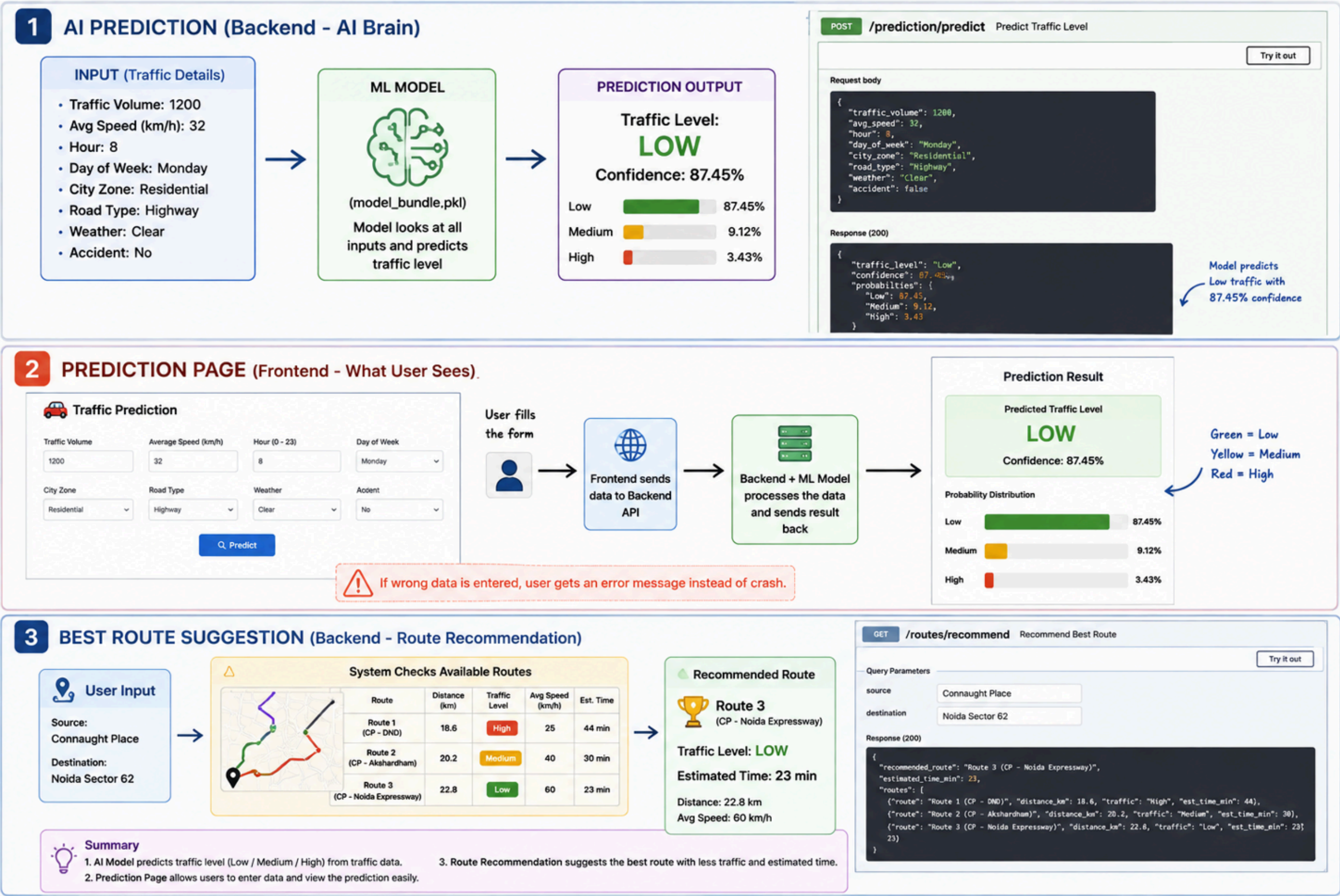
The system provides an alternate route by comparing the congestion levels of different available routes.

Congestion scores:

- Low = 1
- Medium = 2
- High = 3

The routes are compared based on these scores. The route with the lowest congestion score is considered the better route and is recommended.

ROUTE RECOMMENDATION – OVERALL FLOW



PREDICTION PAGE – USER VIEW

Traffic Vision AI - Prediction

← → ↻ http://127.0.0.1:5173/predict

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Traffic Vision AI

Dashboard

Live Map

Traffic Prediction

Best Route

Reports

Alerts

Logout

Team 3

Traffic Prediction

Enter traffic details to predict the traffic level

Enter Traffic Details

Traffic Volume

1200

Average Speed (km/h)

32

Hour (0 - 23)

8

Day of Week

Monday

City Zone

Residential

Road Type

Highway

Weather

Clear

Accident

No

Predict

Please enter all the details correctly and click on Predict to get the traffic prediction.

Prediction Result

Predicted Traffic Level

LOW

Confidence: 87.45%

Probability Distribution

Low

87.45%

Medium

9.12%

High

3.43%

Green = Low

Yellow = Medium

Red = High

